Exercise Set 12.2 Solutions

Question 1: True or false?

(i) A parallelogram with a right angle is a rectangle.

Verdict: **TRUE**

Reasoning: In a parallelogram, opposite angles are equal and consecutive angles sum to 180°. If one angle is 90°, all four interior angles must be 90°, which fulfills the definition of a rectangle.

(ii) A rhombus with perpendicular diagonals is a square.

Verdict: **FALSE**

Reasoning: The diagonals of *every* rhombus are perpendicular to each other. For a rhombus to be a square, its angles must also be right angles (or equivalently, its diagonals must be equal in length).

(iii) If the diagonals of a parallelogram are equal, then it is a rectangle.

Verdict: **TRUE**

Reasoning: A parallelogram with equal diagonals forces all of its interior angles to be 90°, making it a rectangle.

Question 2

Question: The diagonal AC of a parallelogram $ABCD$ bisects $\angle A$. Show that it also bisects $\angle C$ and that $ABCD$ is a rhombus.

Part 1: Show AC bisects $\angle C$

1. Since $ABCD$ is a parallelogram, $AB \parallel CD$ and $AD \parallel BC$.
2. Given that AC bisects $\angle A$:

$$\angle BAC = \angle DAC \quad \text{—— (1)}$$

3. Since $AB \parallel CD$ with transversal AC, alternate interior angles are equal:

$$\angle BAC = \angle DCA \quad \text{—— (2)}$$

4. Since $AD \parallel BC$ with transversal AC, alternate interior angles are equal:

$$\angle DAC = \angle BCA \quad \text{—— (3)}$$

5. From equations (1), (2), and (3), it follows directly that:

$$\angle BCA = \angle DCA$$

6. Hence, AC **bisects** $\angle C$.

Part 2: Show $ABCD$ is a rhombus

1. In ΔABC, since $\angle BAC = \angle BCA$ (from equations 1, 2, 3 above):

$$AB = BC \quad \textit{(sides opposite to equal angles are equal)}$$

2. A parallelogram with adjacent sides equal ($AB = BC$) has all four sides equal:

$$AB = BC = CD = DA$$

3. Therefore, $ABCD$ **is a rhombus**.

Question 3: Converses of True Properties

Answer with Yes or No. If No, state what extra condition makes it Yes.

(i) If the diagonals of a quadrilateral $ABCD$ bisect its angles, must $ABCD$ be a rhombus?

Answer: **YES**

Explanation: If the diagonals bisect the interior angles, the opposite sides are forced to be parallel and all four adjacent sides are equal, defining a rhombus.

(ii) If the diagonals of a quadrilateral $ABCD$ bisect each other at right angles, must $ABCD$ be a rhombus?

Answer: **YES**

Explanation: A quadrilateral whose diagonals bisect each other at 90° is by definition a rhombus.

(iii) If the diagonals of a quadrilateral $ABCD$ are of equal length, must $ABCD$ be a rectangle?

Answer: **NO**

Extra Condition Required: The quadrilateral must first be a **parallelogram** (or the diagonals must also bisect each other). *Note: An isosceles trapezoid has diagonals of equal length, but it is not a rectangle.*

Question 4

Question: Let $ABCD$ be a parallelogram with $AB \neq BC$. Show that the pairwise intersection points of the four angle bisectors form the vertices of a rectangle.

Solution:

Let the angle bisectors of $\angle A, \angle B, \angle C, \angle D$ intersect to form quadrilateral $PQRS$.

1. Consecutive interior angles in a parallelogram sum to 180°:

$$\angle A + \angle D = 180°$$

2. Bisecting these adjacent angles gives:

$$\tfrac{1}{2}\angle A + \tfrac{1}{2}\angle D = 90°$$

3. In ΔADS, applying the angle sum property:

$$\angle DAS + \angle ADS + \angle ASD = 180°$$

$$90° + \angle ASD = 180° \Rightarrow \angle ASD = 90°$$

4. Since vertically opposite angles are equal:

$$\angle PSR = \angle ASD = 90°$$

5. Applying the exact same angle sum property to ΔAPB, ΔBQC, and ΔCRD, we obtain:

$$\angle P = 90°, \quad \angle Q = 90°, \quad \angle R = 90°, \quad \angle S = 90°$$

6. A quadrilateral with four right angles is a rectangle. Therefore, $PQRS$ **is a rectangle**.